

PitchCoach AI

Cost Analysis & Infrastructure Report

Prepared for Automagikal

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Executive Summary

PitchCoach AI is an AI-powered pitch coaching platform designed to help founders and entrepreneurs improve their investor pitches through four comprehensive coaching modules. This document provides a detailed cost analysis for infrastructure, AI services, and operational expenses at scale, with projections for 1,000 active users.

Key Findings: The platform is designed with cost efficiency as a core principle. Using a combination of free tiers during development, pay-as-you-go AI services, and serverless infrastructure, the estimated monthly operational cost for 1,000 active users ranges from \$400 to \$800, with per-session AI costs averaging \$0.15 to \$0.70 depending on the coaching module utilized.

Metric	MVP Stage	1,000 Users
Monthly Infrastructure	\$0 - \$50	\$100 - \$200
Monthly AI Costs	\$10 - \$20	\$300 - \$600
Total Monthly Cost	\$10 - \$70	\$400 - \$800
Cost Per Active User	N/A	\$0.40 - \$0.80

Technology Stack Overview

PitchCoach AI is built on a modern, scalable technology stack optimized for cost efficiency and developer productivity. The architecture leverages serverless computing, managed databases, and specialized AI services to minimize operational overhead while maximizing performance.

Component	Service	Purpose
Frontend	Next.js 16 + React 19	Server-side rendered UI with App Router
Styling	Tailwind CSS + shadcn/ui	Responsive design with component library
Database	Supabase (PostgreSQL)	Primary data storage with RLS security
Authentication	Clerk	OAuth + email authentication
File Storage	Cloudflare R2	PDF/PPTX deck uploads
Video Processing	Cloudflare Stream	Pitch video hosting and processing
AI - Text	Anthropic Claude Sonnet 4	Document and script analysis

AI - Video	Google Gemini	Video analysis and transcription
Billing	Stripe	Subscriptions and one-time purchases
Hosting	Vercel	Edge deployment and CDN

Coaching Modules & AI Costs

PitchCoach AI offers four distinct coaching modules, each leveraging different AI capabilities and incurring varying computational costs. Understanding these per-session costs is critical for pricing strategy and margin analysis.

E1: Pitch Deck Analyser

The Pitch Deck Analyser accepts PDF or PPTX files and provides comprehensive content scoring across eight dimensions, plus a visual audit of design quality. The module uses Claude Sonnet 4 for document parsing and analysis. Typical document processing requires approximately 5,000 input tokens and 2,000 output tokens per session, resulting in a per-session cost of approximately \$0.045. This module has the lowest AI cost due to efficient text processing.

Scoring Dimensions: Problem Clarity, Solution Clarity, Market Opportunity, Business Model, Team Credibility, Traction, Financials, Ask Clarity. Visual Audit: Design Consistency, Readability, Visual Hierarchy, Color Scheme, Typography.

E2: Elevator Pitch Script Coach

The Script Coach analyzes elevator pitch scripts for the five essential elements of a compelling pitch: Hook, Problem, Solution, Credibility, and Call-to-Action. Input can be direct text, PDF, or DOCX upload. With approximately 2,000 input tokens and 1,000 output tokens per analysis, the per-session cost averages \$0.021. The module also provides AI-rewritten versions and alternative opening hooks.

Features: 5-element scoring, tone analysis, estimated duration, word count, AI-generated rewrites, alternative hooks.

E3: Live Elevator Pitch Coach

The Live Pitch module accepts video recordings up to 3 minutes and analyzes both delivery (pace, clarity, filler words, energy, confidence) and body language (eye contact, facial expressions, gestures, posture). Video processing uses Gemini 2.5 Flash for cost efficiency on short-form content. Per-session costs range from \$0.05 to \$0.10 depending on video duration and transcript length.

Features: Delivery scoring, body language analysis, transcript generation, key moment identification, personalized coaching drills.

E4: Full Pitch Session

The most comprehensive module combines deck analysis with video processing for sessions up to 30 minutes. This module provides 6-dimension investor readiness scoring, anticipated Q&A; questions, and prioritized recommendations. Due to the extended video length, this module uses Gemini 1.5 Pro for its larger context window, resulting in higher per-session costs ranging from \$0.50 to \$1.00.

Features: 6-dimension scoring (Problem-Solution Fit, Market Opportunity, Business Model Viability, Team Credibility, Traction/Milestones, Delivery/Presence), investor readiness level assessment, anticipated questions with suggested answers.

Per-Session AI Cost Summary

Module	AI Service	Tokens (Avg)	Cost/Session
E1: Deck Analyser	Claude Sonnet 4	7,000	\$0.05
E2: Script Coach	Claude Sonnet 4	3,000	\$0.02
E3: Live Pitch	Gemini 2.5 Flash	Video + 5,000	\$0.08
E4: Full Session	Claude + Gemini Pro	Video + 20,000	\$0.62

Infrastructure Cost Analysis

This section details the cost structure for each infrastructure component, comparing MVP-stage pricing with projected costs at 1,000 active users. All pricing reflects current provider rates as of March 2026.

Database: Supabase (PostgreSQL)

Supabase provides a managed PostgreSQL database with built-in authentication, real-time subscriptions, and storage. The platform offers generous free tiers ideal for development and early-stage testing, with predictable scaling costs for production workloads.

Tier	Price/Month	Storage	Bandwidth	Best For
Free	\$0	500 MB	5 GB	MVP / Testing
Pro	\$25	8 GB	250 GB	Production
Team	\$599	8 GB+	Custom	Enterprise

Recommendation: Begin with the Free tier during development. Upgrade to Pro (\$25/month) when approaching storage limits or requiring production-grade support. At 1,000 users, the Pro tier provides ample headroom.

Authentication: Clerk

Clerk provides modern authentication with OAuth providers (Google, LinkedIn), email/password, and multi-factor authentication. The free tier supports up to 10,000 monthly active users, making it ideal for startups and growing applications.

Tier	Price/Month	MAUs	Features
Free	\$0	10,000	OAuth, MFA, Webhooks
Pro	\$25+	10,000+	SSO, Advanced fraud protection

Recommendation: The Free tier covers the entire projected user base of 1,000 users with significant headroom. No upgrade needed until exceeding 10,000 MAUs.

File Storage: Cloudflare R2

Cloudflare R2 provides S3-compatible object storage without egress fees, making it ideal for user-uploaded pitch decks (PDF/PPTX files). The pay-as-you-go model ensures costs scale linearly with usage.

Resource	Rate	Est. 1K Users/Month
Storage	\$0.015/GB	\$2 - \$5

Class A Operations (writes)	\$4.50/million	< \$1
Class B Operations (reads)	\$0.36/million	< \$0.50

Recommendation: R2 is highly cost-effective. At 1,000 users with average deck uploads of 5MB each, expect \$5-10/month in storage costs.

Video Processing: Cloudflare Stream

Cloudflare Stream handles video uploads, transcoding, and delivery for the E3 and E4 coaching modules. The service provides automatic optimization and global CDN distribution.

Resource	Rate	Est. 1K Users/Month
Storage	\$5/1,000 minutes	\$10 - \$20
Streaming	\$1/1,000 minutes	\$5 - \$10
Uploads	Free	\$0

Recommendation: For 1,000 users with mixed E3 (3-min) and E4 (30-min) sessions, budget \$15-30/month for video infrastructure.

Hosting: Vercel

Vercel provides edge deployment with automatic SSL, preview deployments, and seamless GitHub integration. The platform is optimized for Next.js applications with global CDN distribution.

Tier	Price/Month	Bandwidth	Build Minutes
Hobby	\$0	100 GB	6,000
Pro	\$20	1 TB	Unlimited
Enterprise	Custom	Custom	Custom

Recommendation: Start with Hobby (free). Upgrade to Pro (\$20/month) at launch for production features, analytics, and priority support.

Billing: Stripe

Stripe handles subscription management, one-time purchases, and bundle sales. Stripe operates on a transaction-fee model with no monthly costs, making it ideal for variable revenue streams.

Transaction Type	Fee Structure
Standard Cards	2.9% + \$0.30 per transaction
International Cards	3.9% + \$0.30 per transaction
Subscription Billing	Included in standard fee

Note: Stripe fees are variable based on revenue. At \$10,000 monthly revenue, expect approximately \$320 in processing fees (3.2% effective rate).

1,000 Active User Cost Projection

This section provides a detailed cost projection for operating PitchCoach AI with 1,000 monthly active users. Assumptions include average session usage patterns and typical user behavior based on industry benchmarks for SaaS coaching platforms.

Usage Assumptions: Each of the 1,000 users averages 2 sessions per month across all modules. Distribution: E1 (40%), E2 (35%), E3 (15%), E4 (10%). This reflects the lower barrier to entry for text-based modules versus video-intensive sessions.

Monthly Cost Breakdown at Scale

Category	Service	Low Estimate	High Estimate
Database	Supabase Pro	\$25	\$25
Authentication	Clerk Free	\$0	\$0
File Storage	Cloudflare R2	\$5	\$10
Video Processing	Cloudflare Stream	\$15	\$30
Hosting	Vercel Pro	\$20	\$20
AI - E1 Sessions (800)	Claude Sonnet 4	\$40	\$50
AI - E2 Sessions (700)	Claude Sonnet 4	\$14	\$20
AI - E3 Sessions (300)	Gemini 2.5 Flash	\$24	\$30
AI - E4 Sessions (200)	Claude + Gemini Pro	\$100	\$200
TOTAL		\$243	\$385

Key Financial Metrics

Metric	Value	Notes
Cost Per Active User	\$0.24 - \$0.39	Infrastructure + AI combined
Cost Per Session	\$0.12 - \$0.19	Average across all modules
AI Cost Ratio	60-70%	AI represents majority of variable costs
Infrastructure Cost Ratio	30-40%	Fixed costs for database, hosting
Breakeven Revenue/User	\$1.00 - \$1.50	Assuming 80% gross margin target

Scaling Considerations & Cost Optimization

As PitchCoach AI grows beyond 1,000 users, several strategies can help manage costs while maintaining service quality. This section outlines key optimization opportunities and projected costs at higher scales.

Cost Optimization Strategies

- 1. Response Caching: Implement caching for repeated analyses of similar content. If a user re-uploads the same deck with minor changes, leverage cached analysis results. Expected savings: 15-20% on AI costs.
- 2. Smart Video Processing: For E4 sessions, implement a two-stage analysis. First, use Gemini Flash for initial assessment, only escalating to Gemini Pro for sessions flagged as needing deeper analysis. Expected savings: 30-40% on E4 AI costs.
- 3. Batch Processing: Queue non-urgent analyses for off-peak processing. This enables bulk API usage and potential volume discounts with AI providers.
- 4. Tiered Access: Limit high-cost modules (E4) to premium subscriptions. Structure pricing to ensure profitable unit economics per module.

Scale Projections

Users	Monthly Cost	Cost/User	Recommended Actions
100	\$50 - \$100	\$0.50 - \$1.00	Free tiers, development focus
1,000	\$250 - \$400	\$0.25 - \$0.40	Pro tiers, optimize AI usage
5,000	\$800 - \$1,500	\$0.16 - \$0.30	Volume discounts, caching
10,000	\$1,500 - \$2,500	\$0.15 - \$0.25	Enterprise DB tier, CDN optimization
25,000	\$3,000 - \$5,000	\$0.12 - \$0.20	Custom AI agreements, dedicated infra

Risk Factors & Considerations

Investors should be aware of the following risk factors that could impact cost projections and operational stability.

AI Pricing Volatility: The AI services market is rapidly evolving. While current pricing from Anthropic and Google is competitive, price changes could significantly impact unit economics. Mitigation: Maintain relationships with multiple AI providers and build abstraction layers for easy model switching.

Usage Pattern Variance: Actual user behavior may differ from projections. Higher-than-expected E4 usage would increase costs disproportionately. Mitigation: Implement usage monitoring and dynamic pricing adjustments.

Service Dependencies: The platform relies on multiple third-party services (Supabase, Clerk, Cloudflare). Service outages or policy changes could impact operations. Mitigation: Maintain disaster recovery procedures and consider multi-region deployment for critical services.

Video Processing Limits: Extended video sessions (E4) consume significant AI resources. Unexpected growth in this module could strain budgets. Mitigation: Implement strict session duration limits and tiered access controls.

Conclusion & Recommendations

PitchCoach AI is positioned for cost-effective scaling with a well-architected technology stack. The estimated monthly operational cost for 1,000 active users ranges from \$250 to \$400, with per-user costs decreasing as scale increases due to infrastructure efficiency gains.

Key Recommendations:

- Start with free tiers during MVP development to minimize burn rate while validating product-market fit.
- Implement robust usage monitoring before launch to track AI costs per module and identify optimization opportunities.
- Structure pricing tiers to ensure profitable unit economics, with higher-cost modules (E4) reserved for premium subscribers.
- Build caching infrastructure early to reduce redundant AI API calls and improve response times.
- Maintain flexibility in AI provider selection through abstraction layers to capitalize on pricing competition.

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